Language of Sequences & Series

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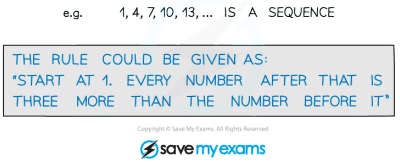
Language of sequences & series

What is a sequence?

- A **sequence** is an ordered set of numbers with a rule for finding all the numbers in the sequence



- The numbers in a sequence are called **terms**
- The terms of a sequence are often referred to by letters with a subscript

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

THE GENERAL TERM OF THE SEQUENCE WOULD BE

$u_n = 3n - 2$

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What is a series?

- You get a **series** by summing up the terms in a sequence

e.g. FOR THE SEQUENCE 1, 4, 7, 10, 13, ...

THE ASSOCIATED SERIES IS $1 + 4 + 7 + 10 + 13 + \ldots$

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- We use the notation S_n to refer to the sum of the first n terms in the series
  i.e. $S_n = u_1 + u_2 + u_3 + \ldots + u_n$

SO FOR THE SERIES ABOVE

$S_1 = 1$

$S_2 = 1 + 4 = 5$

$S_3 = 1 + 4 + 7 = 12$

$S_4 = 1 + 4 + 7 + 10 = 22$

$S_5 = 1 + 4 + 7 + 10 + 13 = 35$

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What are increasing, decreasing and periodic sequences?

- A sequence is **increasing** if $u_{n+1} > u_n$ for all positive integers n
  - i.e. if every term is greater than the term before it
- A sequence is **decreasing** if $u_{n+1} < u_n$ for all positive integers n
  - i.e. if every term is less than the term before it

e.g.

3, 7, 11, 15, 19, ... IS AN INCREASING SEQUENCE

64, 32, 16, 8, 4, ... IS A DECREASING SEQUENCE

1, −2, 3, −4, 5, ... IS NEITHER AN INCREASING NOR A DECREASING SEQUENCE

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- A sequence is **periodic** if the terms repeat in a cycle
- The **order** (or **period**) of a periodic sequence is the number of terms in each repeating cycle

e.g.

1, 2, 3, 1, 2, 3, 1, 2, 3, ... IS A PERIODIC SEQUENCE OF ORDER 3

2, −2, 2, −2, 2, −2, ... IS A PERIODIC SEQUENCE OF ORDER 2

0, 0, 0, 0, 0, 0, 0, ... IS A PERIODIC SEQUENCE OF ORDER 1

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Examiner Tips and Tricks

Look out for sequences defined by trigonometric functions – this can be a way of 'hiding' a periodic function.

$u_n = \cos(180n^\circ)$

$u_1 = \cos(180^\circ) = -1$
$u_2 = \cos(360^\circ) = 1$
$u_3 = \cos(540^\circ) = -1$
$u_4 = \cos(720^\circ) = 1$

THIS IS THE SEQUENCE $-1, 1, -1, 1, -1, 1, \ldots$
(i.e. A PERIODIC SEQUENCE OF ORDER 2)

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Worked Example

Determine the order of the following periodic functions

a) $u_n = 3 + (-1)^n$

b) $u_n = \sin(90n^\circ)$

WORK OUT THE TERMS AND LOOK FOR A PATTERN

a) $u_1 = 3 + (-1)^1 = 3 - 1 = 2$
$u_2 = 3 + (-1)^2 = 3 + 1 = 4$
$u_3 = 3 + (-1)^3 = 3 - 1 = 2$
$u_4 = 3 + (-1)^4 = 3 + 1 = 4$

THE SEQUENCE IS $2, 4, 2, 4, 2, 4, \ldots$

PERIODIC WITH ORDER 2

b) $u_1 = \sin 90^\circ = 1$ $\quad u_2 = \sin 180^\circ = 0$
$u_3 = \sin 270^\circ = -1$ $\quad u_4 = \sin 360^\circ = 0$
$u_5 = \sin 450^\circ = 1$ $\quad u_6 = \sin 540^\circ = 0$
$u_7 = \sin 630^\circ = -1$ $\quad u_8 = \sin 720^\circ = 0$

SEQUENCE IS $1, 0, -1, 0, 1, 0, -1, 0, \ldots$

PERIODIC WITH ORDER 4

BE CAREFUL HERE. EVERY SECOND TERM IS A ZERO, BUT IT TAKES FOUR TERMS FOR THE PATTERN TO REPEAT

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